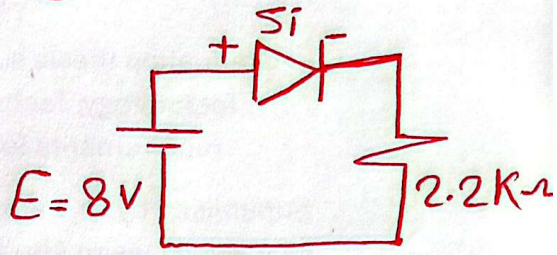
Diode Applications

Ex①: for the series diode Configwration Determine V_D, V_R and I_D?

Sol

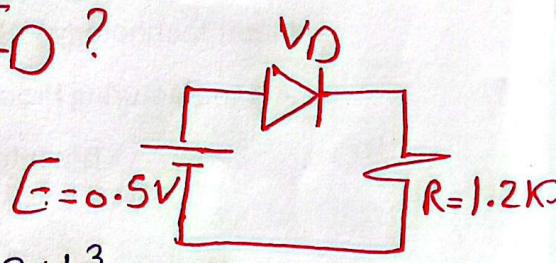
$V_D = 0.7\,V$



$V_R = E - V_D = 8 - 0.7 = 7.3V$

$I_D = I_R = \frac{V_R}{R} = \frac{7.3}{2.2\times10^3} = 3.32\,mA$

Ex②: For the series diode Configwration Determine V_D, V_R and I_D?

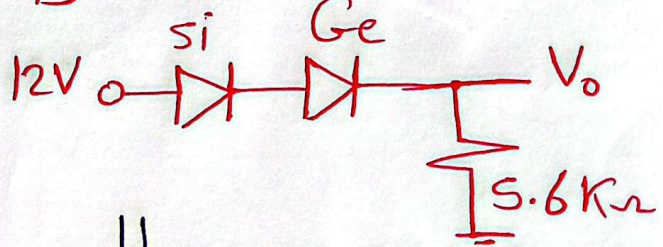


$I_D = 0A$

$V_R = I_R R = I_D R = 0 * 1.2*10^3 = 0V$

$V_D = E = 0.5V$

Ex③: Determine V_o and I_D for the series cir.



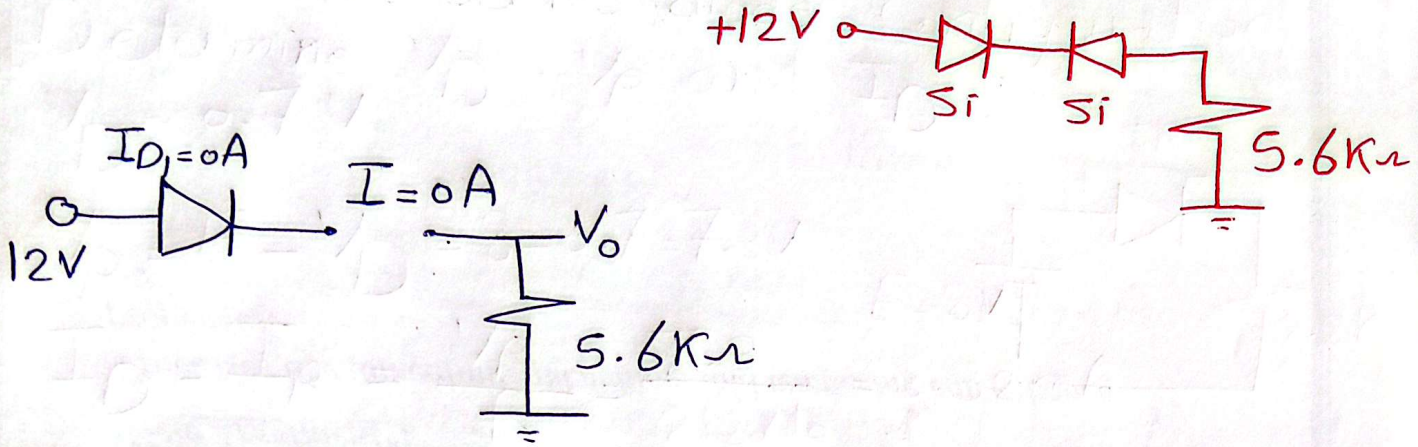
$V_o = E - V_{Si} - V_{Ge}$

$= 12 - 0.7 - 0.3 = 11V$

$I_D = I_R = \frac{V_o}{R} = \frac{11}{5.6K} = \frac{11}{5.6*10^3} = 1.96\,mA$

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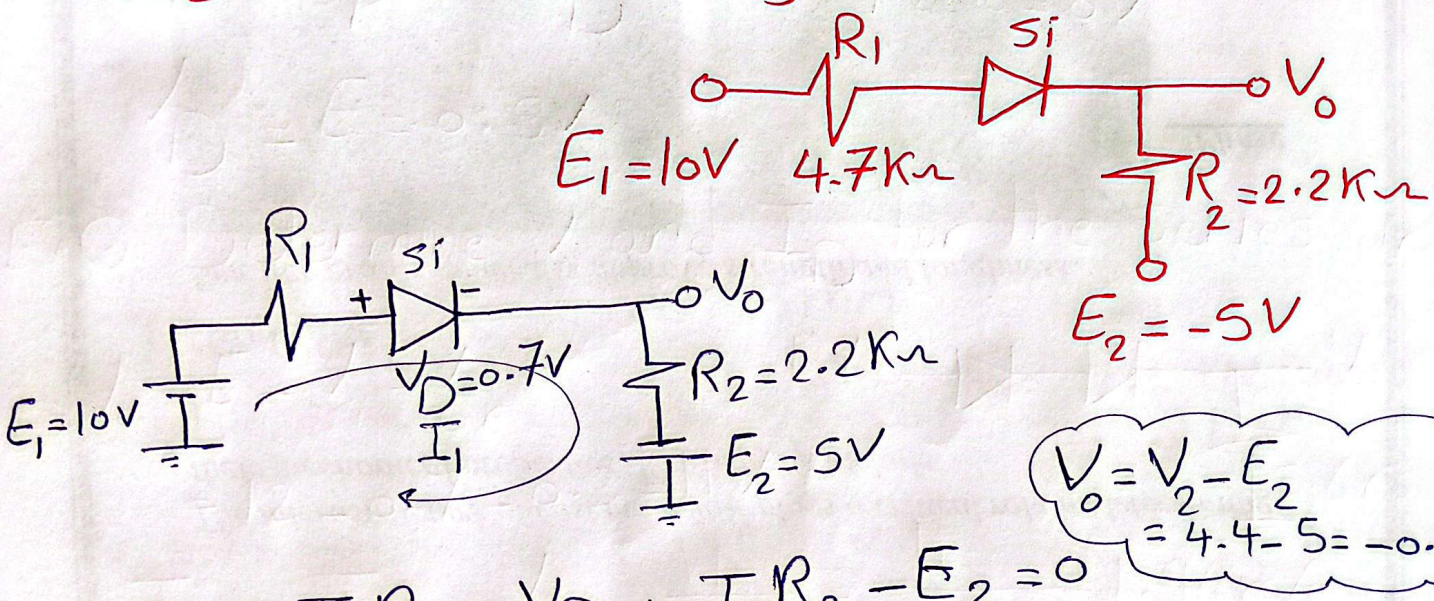
Ex④: Determine I_D, V_D for the circuit



$V_o = I * R = 0 * 5.6K = 0\ V$

$V_{D_2} = 12\ V$

Ex⑤: Determine I, V_1, V_2 and V_o for the series dc Configuration



$V_o = V_2 - E_2$
$= 4.4 - 5 = -0.6V$

$-E_1 + I_1R_1 + V_D + I_1R_2 - E_2 = 0$

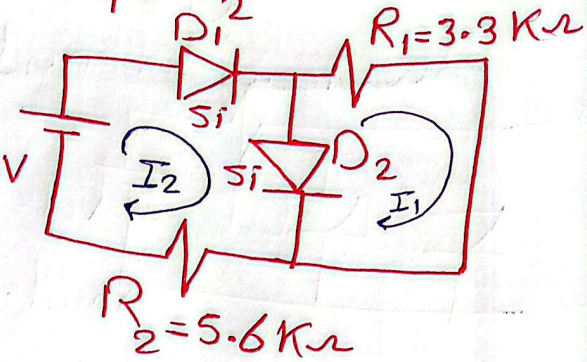
$-10 + 4.7K I_1 + 0.7 + 2.2K I_1 - 5 = 0$

$I_1 = \frac{10 - 0.7 + 5}{4.7K + 2.2K} = 2\ mA$

$V_1 = I_1R_1 = 2\times10^{-3}\times4.7\times10^{3} = 9.4V$

$V_2 = I_1R_2 = 2m\times2.2K = 4.4V$ (2)

Ex⑨: Determine the Currents I_1, I_2 and I_{D_2} for the network?



$$I_{R_1} = \frac{V_{D_2}}{R_1} = \frac{0.7}{3.3K} = 0.212\,mA$$

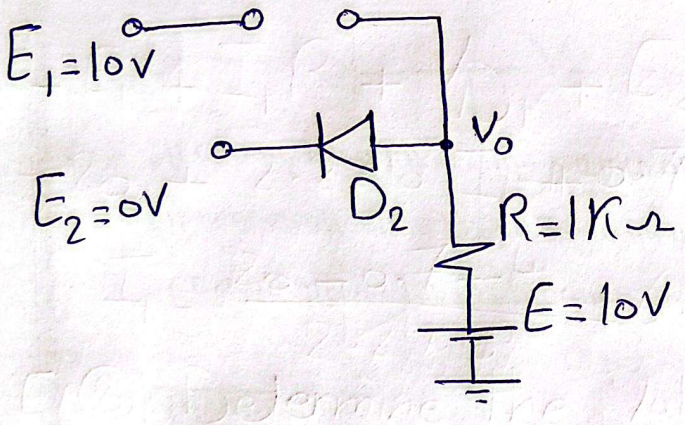
$$-E + V_{D_1} + V_{D_2} + I_2 R_2 = 0$$

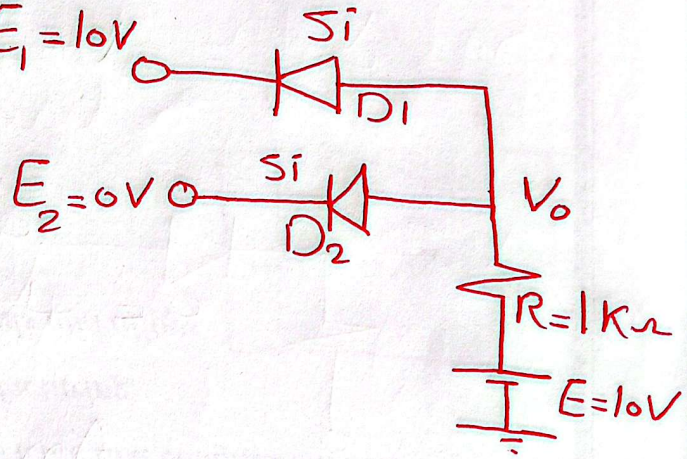
$$-20 + 0.7 + 0.7 + 5.6\times10^{3} I_2 = 0$$

$$I_2 = \frac{20 - 0.7 - 0.7}{5.6\times10^{3}} = 3.32\,mA$$

$$I_{D_2} = I_2 - I_1 = 3.32 - 0.212 = 3.108\,mA$$

Ex⑩: Determine V_o ?

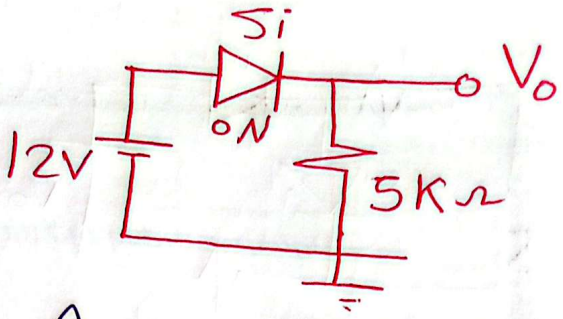




$$I_R = \frac{E - V_{D_2}}{R} = \frac{10 - 0.7}{1K} = 9.3\,mA$$

$$V_o = V_{D_2} = 0.7\,V$$

Ex ⑪: Determine I_R & V_o ?

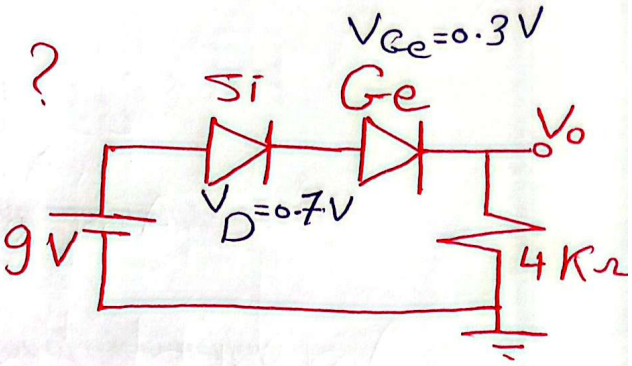


$V_D = 0.7V$

$V_o = 12 - 0.7 = 11.3$

$I_R = \frac{V_o}{R} = \frac{11.3}{5K} = 2.26mA$

Ex ⑫: Determine I_R & V_o ?



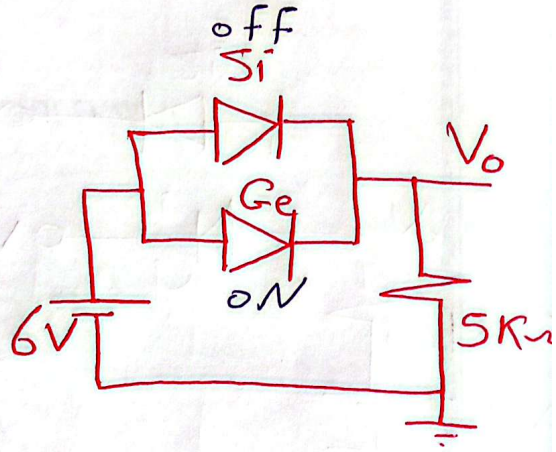
$V_o = E - V_{Si} - V_{Ge}$

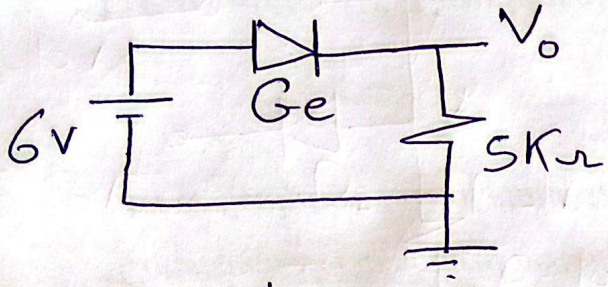
$= 9 - 0.7 - 0.3$

$= 8V$

$I_R = \frac{V_o}{R} = \frac{8}{4K} = 2mA$

Ex ⑬: Determine I_R & V_o ?

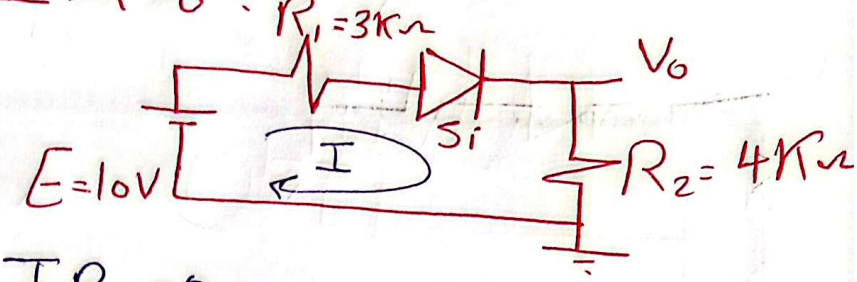




$V_o = E - V_{Ge} = 6 - 0.3 = 5.7V$

$I_R = \frac{V_o}{R} = \frac{5.7}{5K} = 1.14mA$

Ex(14): Determine I & V_o?



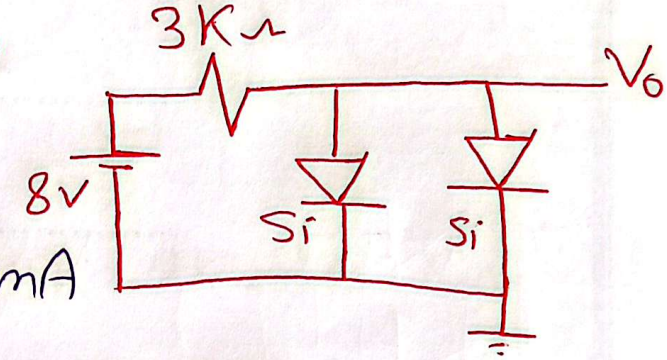
$$-E + IR_1 + V_D + IR_2 = 0$$

$$-10 + 3KI + 0.7 + 4KI = 0$$

$$I = \frac{10 - 0.7}{(3+4)K} = \frac{9.3}{7K} = 1.33\ mA$$

$$V_o = IR_2 = 1.33 * 10^{-3} * 4 * 10^{3} = 5.32\ V$$

Ex(15): Determine V_o?

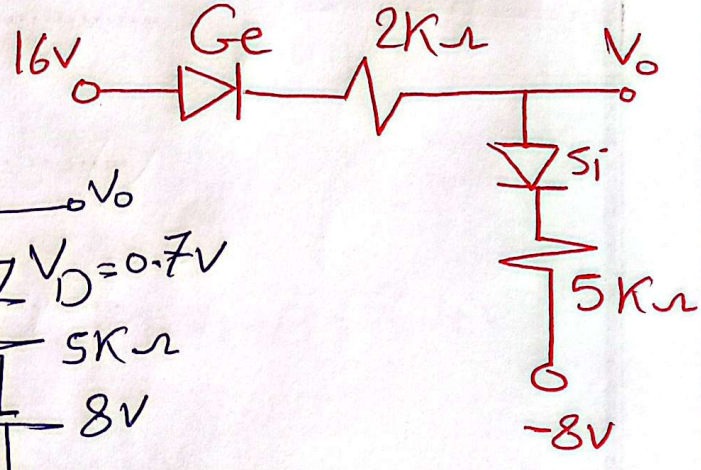


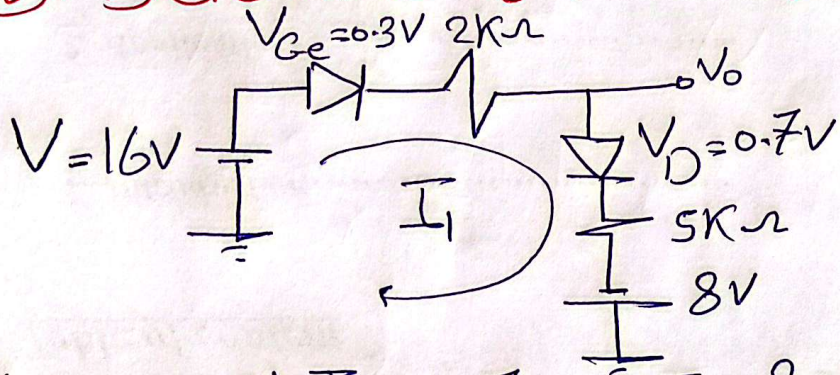
$V_o = 0.7V$

$$I_R = \frac{V}{R} = \frac{8 - 0.7}{3K} = 2.43\ mA$$

$$I_{D_1} = I_{D_2} = \frac{I}{2} = \frac{I_R}{2} = \frac{2.43}{2} = 1.215\ mA$$

Ex(16): Determine V_o?





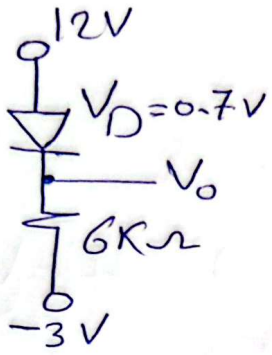
$$-16 + 0.3 + 2KI_1 + 0.7 + 5KI_1 - 8 = 0$$

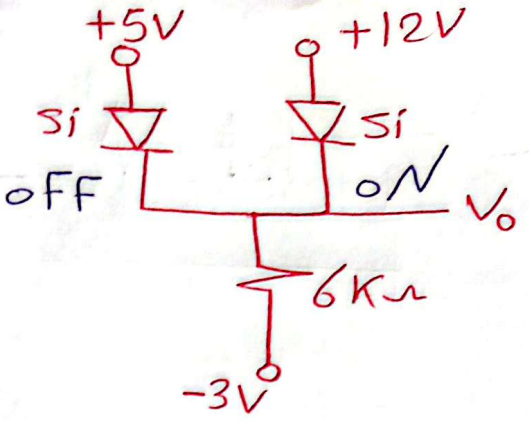
$$I_1 = \frac{16 - 0.3 - 0.7 + 8}{(2+5)K} = 3.29\ mA$$

$$V_o = 16 - 0.3 - (3.29m)(2K) = 9.128\ V$$

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Ex (17) Find V_o ?





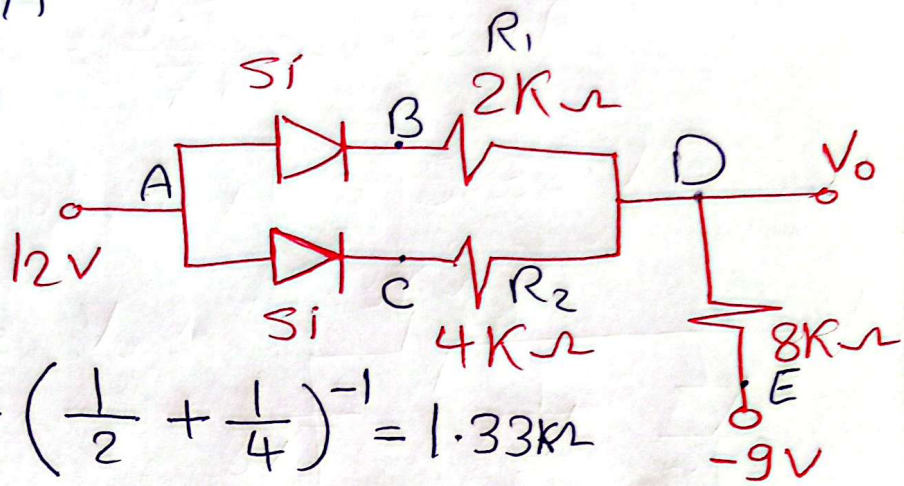
$V_o = 12 - 0.7$

$= 11.3V$

$I_R = \frac{11.3 - (-3)}{6K}$

$= \frac{14.3}{6K} = 2.38mA$

Ex (18): Find V_o ?



$R_{eq} = \left(\frac{1}{R_1} + \frac{1}{R_2}\right)^{-1} = \left(\frac{1}{2} + \frac{1}{4}\right)^{-1} = 1.33K\Omega$

$R_T = R_{eq} + 8K\Omega = 1.33K + 8K = 9.33K\Omega$

$I = \frac{\Delta V}{R_T} = \frac{11.3 - (-9)}{9.33} = 2.18mA$

$\frac{I}{R_1} = 2.18 \frac{4K}{4K+2K} = 1.45mA$

$I_{R_2} = 0.73mA$

$V_o = V_B - I_{R_1} R_1 = 11.3 - 2K * 1.45mA = 8.4V$

$11.3 - (4K * 0.73mA) = 8.38V$

$-9V + (2.18)(8) = 8.4V$